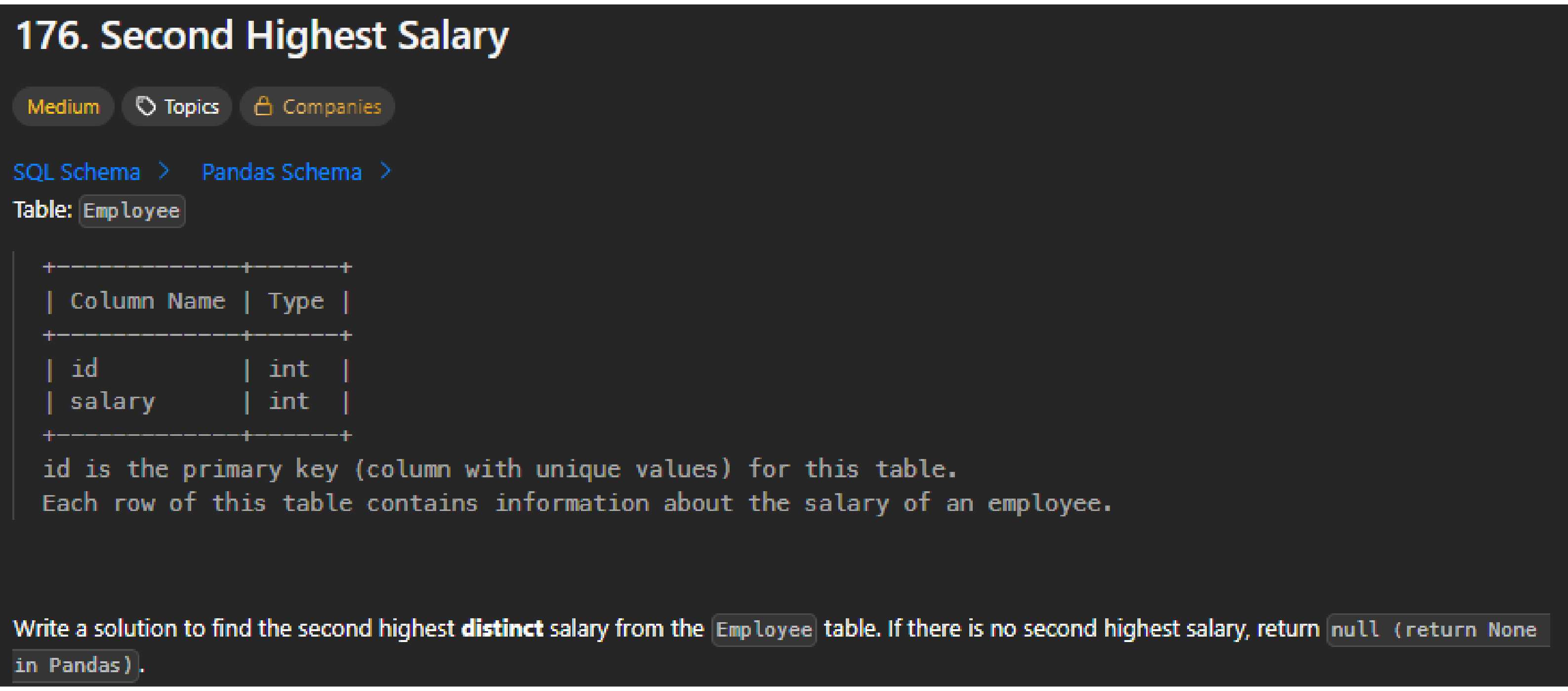
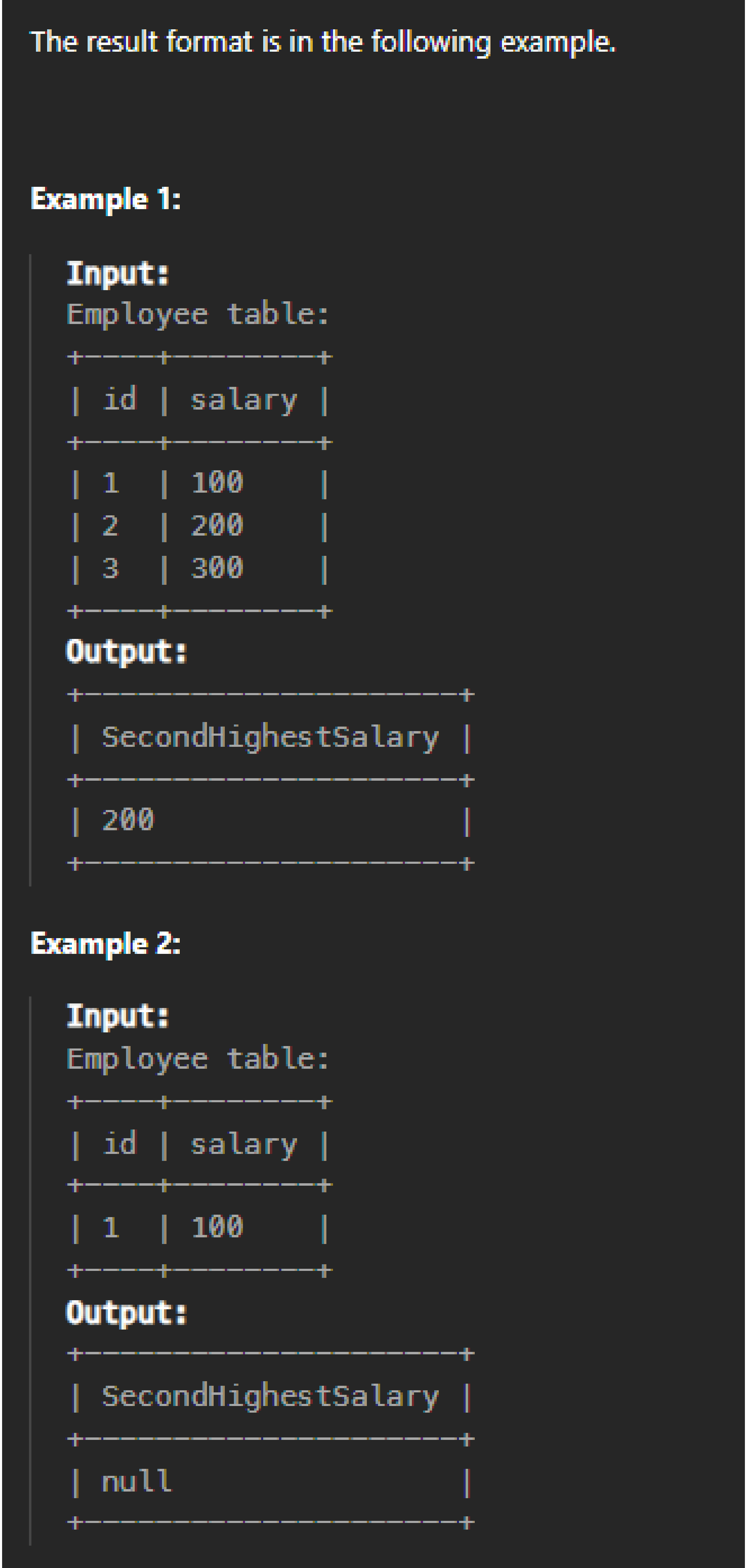


176. Second Highest Salary

Table: `Employee`

```
+-------------+------+
| Column Name | Type |
+-------------+------+
| id          | int  |
| salary      | int  |
+-------------+------+
id is the primary key (column with unique values) for this table.
Each row of this table contains information about the salary of an employee.
```

Write a solution to find the second highest **distinct** salary from the `Employee` table. If there is no second highest salary, return `null (return None in Pandas)`.



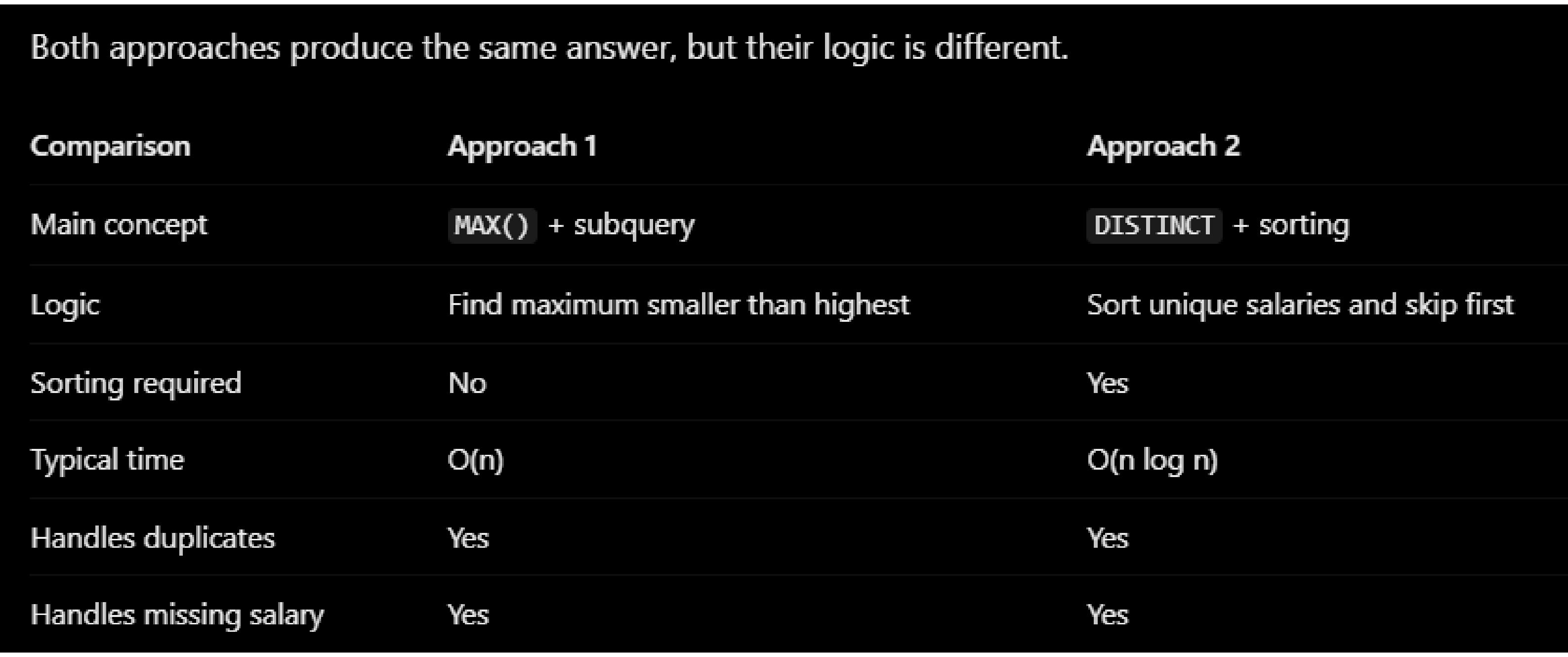
The result format is in the following example.

Example 1:

> **Input:**
> ```
> Employee table:
> +----+--------+
> | id | salary |
> +----+--------+
> | 1  | 100    |
> | 2  | 200    |
> | 3  | 300    |
> +----+--------+
> ```
> **Output:**
> ```
> +---------------------+
> | SecondHighestSalary |
> +---------------------+
> | 200                 |
> +---------------------+
> ```

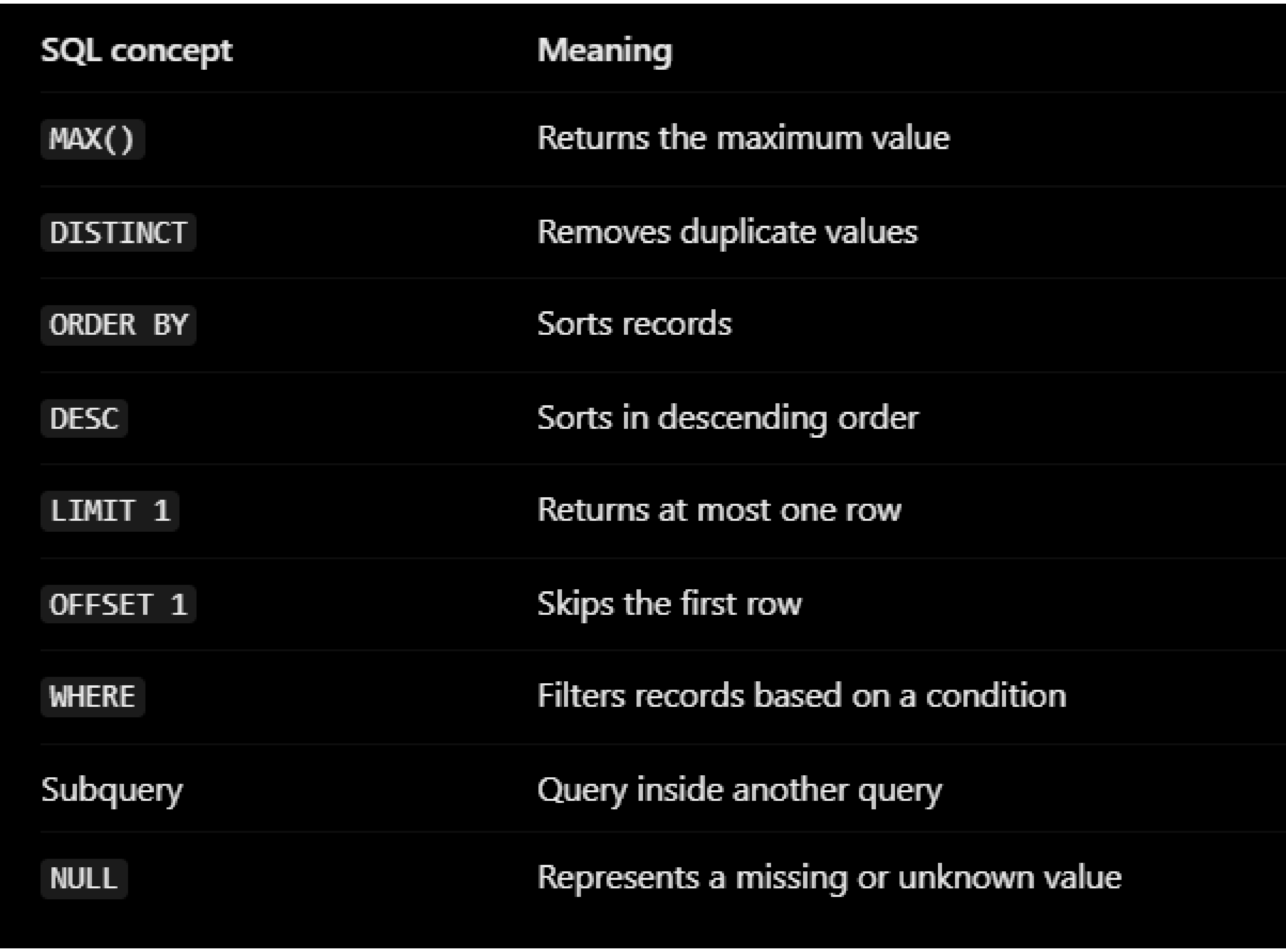
Example 2:

> **Input:**
> ```
> Employee table:
> +----+--------+
> | id | salary |
> +----+--------+
> | 1  | 100    |
> +----+--------+
> ```
> **Output:**
> ```
> +---------------------+
> | SecondHighestSalary |
> +---------------------+
> | null                |
> +---------------------+
> ```

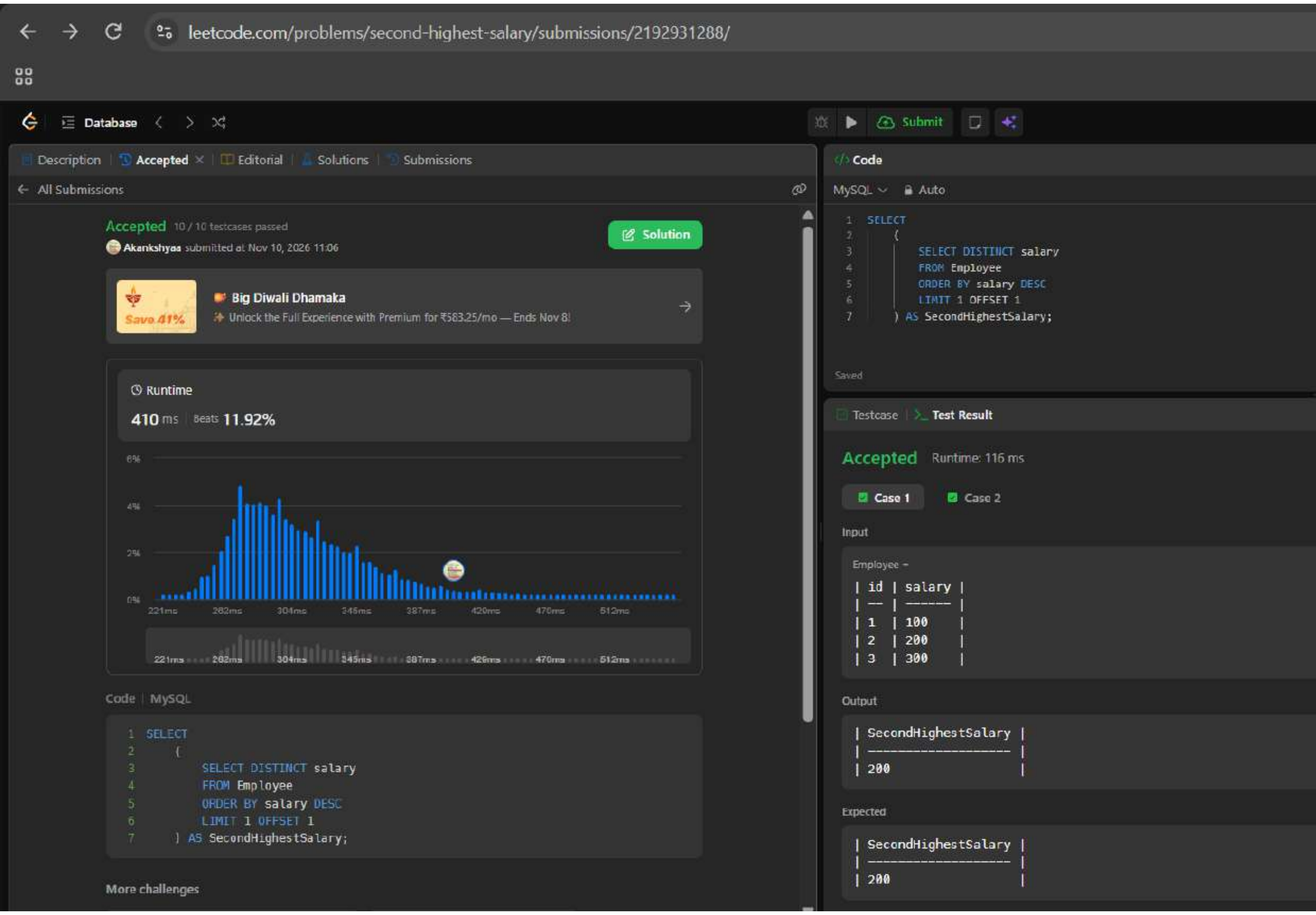


Both approaches produce the same answer, but their logic is different.

Comparison	**Approach 1**	**Approach 2**
Main concept	`MAX()` + subquery	`DISTINCT` + sorting
Logic	Find maximum smaller than highest	Sort unique salaries and skip first
Sorting required	No	Yes
Typical time	O(n)	O(n log n)
Handles duplicates	Yes	Yes
Handles missing salary	Yes	Yes



SQL concept	Meaning
`MAX()`	Returns the maximum value
`DISTINCT`	Removes duplicate values
`ORDER BY`	Sorts records
`DESC`	Sorts in descending order
`LIMIT 1`	Returns at most one row
`OFFSET 1`	Skips the first row
`WHERE`	Filters records based on a condition
Subquery	Query inside another query
`NULL`	Represents a missing or unknown value



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Runtime

410 ms | Beats **11.92%**

Code | MySQL

```sql
SELECT
    (
        SELECT DISTINCT salary
        FROM Employee
        ORDER BY salary DESC
        LIMIT 1 OFFSET 1
    ) AS SecondHighestSalary;
```

```sql
SELECT
    (
        SELECT DISTINCT salary
        FROM Employee
        ORDER BY salary DESC
        LIMIT 1 OFFSET 1
    ) AS SecondHighestSalary;
```

Accepted Runtime: 116 ms

☑ Case 1 ☑ Case 2

Input

Employee =

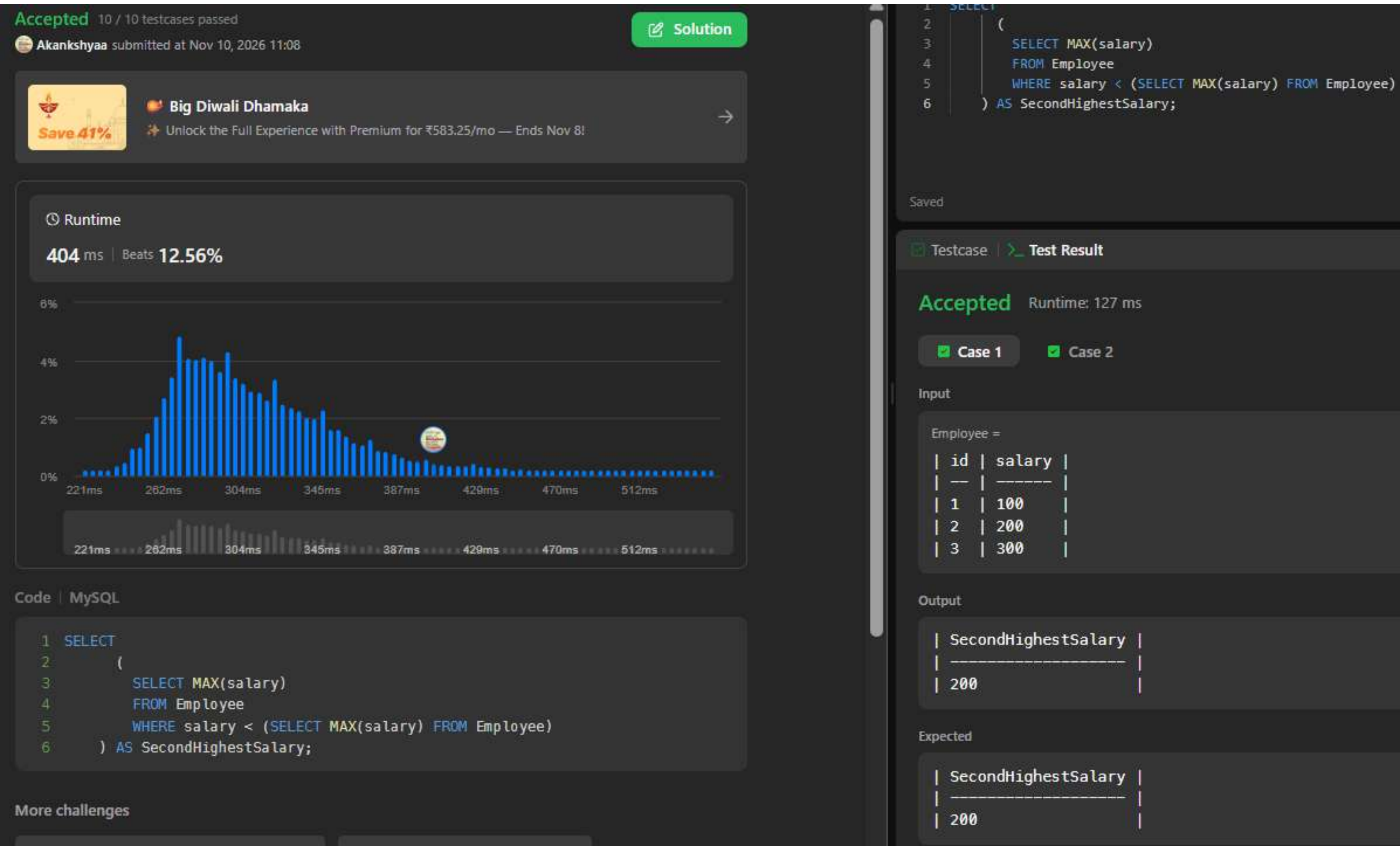
```
| id | salary |
| -- | ------ |
| 1  | 100    |
| 2  | 200    |
| 3  | 300    |
```

Output

```
| SecondHighestSalary |
| ------------------- |
| 200                 |
```

Expected

```
| SecondHighestSalary |
| ------------------- |
| 200                 |
```



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Runtime

404 ms | Beats 12.56%

Code | MySQL

```sql
SELECT
    (
      SELECT MAX(salary)
      FROM Employee
      WHERE salary < (SELECT MAX(salary) FROM Employee)
  ) AS SecondHighestSalary;
```

More challenges

```sql
    (
      SELECT MAX(salary)
      FROM Employee
      WHERE salary < (SELECT MAX(salary) FROM Employee)
  ) AS SecondHighestSalary;
```

Saved

Accepted Runtime: 127 ms

Input

```
Employee =
| id | salary |
| -- | ------ |
| 1  | 100    |
| 2  | 200    |
| 3  | 300    |
```

Output

```
| SecondHighestSalary |
| ------------------- |
| 200                 |
```

Expected

```
| SecondHighestSalary |
| ------------------- |
| 200                 |
```

Ds / Day-12 / lcdbms 10/11/26

leetcode - 176 = Second highest salary

We are given one table, Employee,

id	salary
1	100
2	200
3	300

Our task is to find the 2nd-highest DISTINCT salary.

The word DISTINCT is extremely imp here, it means we must consider only unique salary values, for the above ~~salaries~~ table,

highest salary = 300

2nd " " = 200.

O/P = 200.

but what if the table looks like this?

id	salary
1	100
2	100
3	200
4	200
5	300

the distinct salaries are:- 100, 200 & 300,

highest = 300

2nd " = 200.

we don't count duplicate salaries as separate rankings. What if there is only one employee?

id	salary
1	100

there is no second-highest distinct salary, so we must return NULL.

Approach 1 → using MAX() & subqueries

creates an sql function

~~CREATE FUNCTION getSecondHighestSalary (N INT) RETURNS~~ ~~whatever value the query produces will be returned by the function.~~

~~INT~~

~~starts the function~~ ~~BEGIN~~

~~body~~ SELECT (← finds the max. salary from the filtered employee records.

```
SELECT MAX(salary)
FROM Employee
WHERE salary < (SELECT max(salary)
FROM Employee)
) AS SecondHighestSalary;
```

the inner query finds the highest salary, the outer query considers only salaries smaller than that value.

~~ends the query & function.~~

Approach 2 → Using DISTINCT, ORDER BY, LIMIT & OFFSET

~~CREATE FUNCTION getSecondHighestSalary (N INT)~~

```
SELECT ( SELECT DISTINCT salary
         FROM Employee
         ORDER BY salary DESC
         LIMIT 1 OFFSET 1
       ) AS
SecondHighest salary;
```

→ removes duplicate salary value.

→ ORDER BY sorts the records, DESC means descending order (largest to smallest).

→ returns only one row

↳ skips the 1st row & fetch the next row.